

1.3 Exchanging data	
How data is exchanged between different systems	
1.3.1 Compression, Encryption and Hashing	(a) Lossy vs Lossless compression. (b) Run length encoding and dictionary coding for lossless compression. (c) Symmetric and asymmetric encryption. (d) Different uses of hashing.
1.3.2 Databases	(a) Relational database, flat file, primary key, foreign key, secondary key, entity relationship modelling, normalisation and indexing. See appendix 5f. (b) Methods of capturing, selecting, managing and exchanging data. (c) Normalisation to 3NF. (d) SQL – Interpret and modify. See appendix 5d. (e) Referential integrity. (f) Transaction processing, ACID (Atomicity, Consistency, Isolation, Durability), record locking and redundancy.
1.3.3 Networks	(a) Characteristics of networks and the importance of protocols and standards. (b) The internet structure: • The TCP/IP Stack. • DNS • Protocol layering. • LANs and WANs. • Packet and circuit switching. (c) Network security and threats, use of firewalls, proxies and encryption. (d) Network hardware. (e) Client-server and peer to peer.
1.3.4 Web Technologies	(a) HTML, CSS and JavaScript. See appendix 5d. (b) Search engine indexing. (c) PageRank algorithm. (d) Server and client side processing.